Pulse

A plug-in brain for every small factory machine in India.



Kavya Raval

THE QUESTION

“How much did this machine produce today — and why did it stop?”

Most small-factory owners can't answer. Not because they don't care — because the data doesn't exist.

THE PROBLEM

60M+

MSMEs in India — running on paper registers and guesswork

Idle time

Machines wait, nobody logs it

Micro-stoppages

2-minute halts add up to hours

Silent defects

Found at dispatch, not at source

Industrial monitoring software exists — but it's priced for big plants.

THE IDEA

Clip on. Plug in. See everything.

A low-cost sensor kit that retrofits onto any existing machine — no rewiring, no new machinery.

Rotary encoder

Counts output & speed

Vibration sensor

Flags defect patterns

ESP32 module

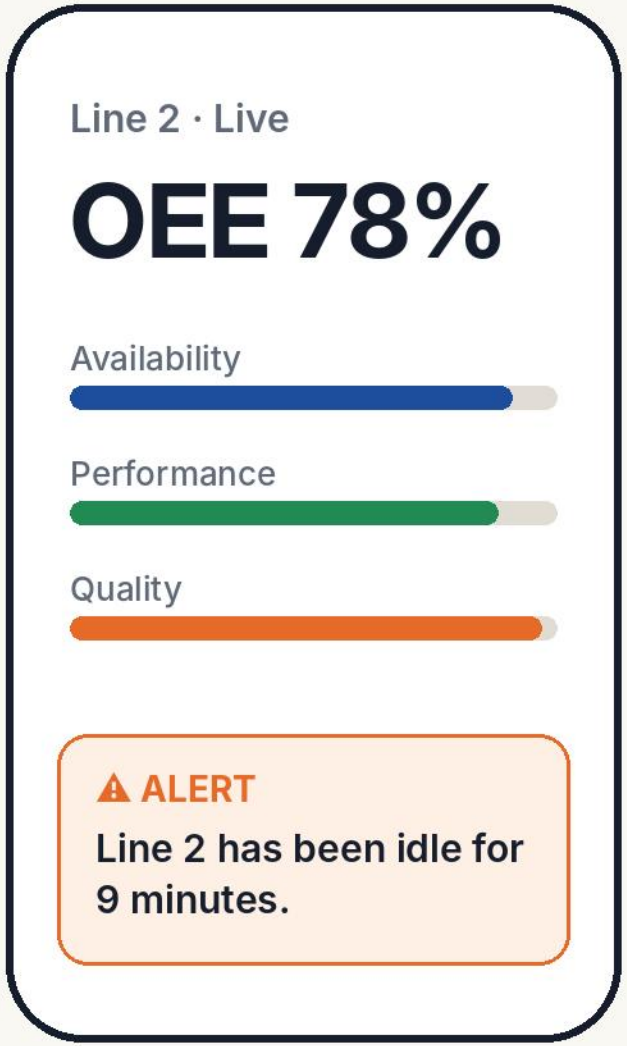
Streams data every second

Target cost: less than a smartphone.

HOW IT WORKS

From machine to phone, every second.

1. Sensor reads the machine
2. Live dashboard computes OEE
3. Plain-language alerts



ALREADY BUILT

Not just a slide.

A working prototype, built for a real wire-fence manufacturer (Duraknot).

✓ Firmware

ESP32 sketch streaming JSON at 1 Hz

✓ Backend

Flask REST API, SQLite, alert engine

✓ Dashboard

Live OEE, deployed online, runs offline too

✓ Tested

Unit tests + benchmark-calibrated simulator

duraknotdashboard.vercel.app

WHY IT MATTERS

+5%

utilisation gain — more output from the machines a factory already owns.

More income
for owners

More jobs
as small units grow

Less waste
of material & energy

Multiply across millions of MSMEs → a stronger backbone for India's economy.

When every machine has a pulse, every maker can grow.



Thank you · Kavya Raval